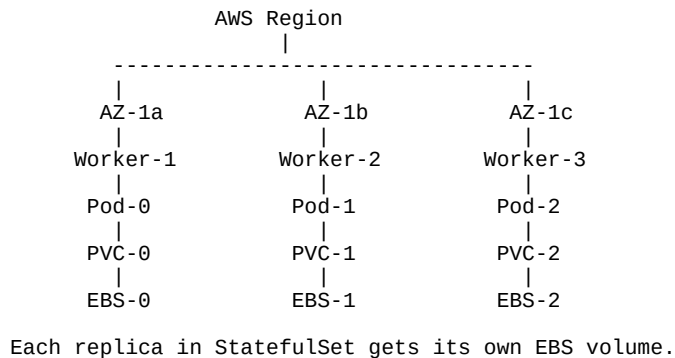


StatefulSet + Amazon EKS Real World Setup Guide

1. Real World Architecture Overview



2. Prerequisites

- EKS Cluster created
- Worker nodes in multiple AZs
- IAM OIDC provider enabled
- AWS EBS CSI Driver installed

3. Install AWS EBS CSI Driver (EKS Add-on)

```
aws eks create-addon --cluster-name my-cluster --addon-name aws-ebs-csi-driver
```

4. Create StorageClass

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ebs-sc
provisioner: ebs.csi.aws.com
volumeBindingMode: WaitForFirstConsumer
parameters:
  type: gp3
  encrypted: "true"
```

5. StatefulSet Example (MySQL)

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mysql
```

```

spec:
  serviceName: mysql
  replicas: 3
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
        - name: mysql
          image: mysql:8
          volumeMounts:
            - name: mysql-data
              mountPath: /var/lib/mysql
  volumeClaimTemplates:
    - metadata:
        name: mysql-data
      spec:
        accessModes: ["ReadWriteOnce"]
        storageClassName: ebs-sc
        resources:
          requests:
            storage: 20Gi

```

6. Production Best Practices

- Use Multi-AZ worker nodes
- Use WaitForFirstConsumer binding mode
- Enable EBS encryption
- Take regular EBS snapshots
- Use PodDisruptionBudgets
- Monitor using CloudWatch + Prometheus